1. Look at the table. The manager is trying to use it to understand how his goal scorers perform.

Player	Goals	Shots	Strike rate
Fergus	8	10	80%
Sophie	6	10	
Louis	26	100	
Nadia	4	5	
Ella	14	20	
Rob	18	25	

a. What is Ella's strike rate (how many goals she scored compared to the shots she takes) expressed as a percentage? _____

b. What is Louis's strike rate expressed as a percentage? _____

c. What is Nadia's strike rate expressed as a percentage? _____

d. What is the percentage difference between Rob's strike rate and Sophie's strike rate? _____

e. Who has the highest strike rate? _____

---

2. Answer the following calculation questions.

a. 3/4 x 5/6 = _____

b. 2/3 – 1/4 = _____

c. 1/5 x 5/8 = _____

d. 3/4 + 1/5 = _____

e. 6.94 + 3.46 = _____

f. 2.16 ÷ 4 = _____

g. 8.25 – 4.19 = _____

h. 2.48 x 5 = _____

i. I start to read at 9.15pm. Write this time using the 24 Hour clock. _____

j. Sunrise is at half past six in the morning. Write this time using the 12 Hour clock. _____

k. If it takes me 35 minutes to walk from my office to a meeting what time do I get to the meeting if I leave the office at 3.50pm? Give your answer using the 24 Hour clock. _____

3. In this number box every row, column and diagonal must add up to 12. Number 0 – 8 inclusive may be used once only.

7	a.	b.
c.	4	d.
3	e.	f.

a. _____ **b.** _____ **c.** _____

d. _____ **e.** _____ **f.** _____

- -

4. Find the lowest common multiple of the following numbers.

Example: The lowest common multiple of 3 and 5 is 15

a. 3 and 4 = _____ **c.** 5 and 7 = _____

b. 7 and 8 = _____ **d.** 6 and 8 = _____

5. On a timetable the following train arrival times are shown. Train A at 09:30 hours, Train B at 11:45 hours and Train C at 15:15 hours.

Put these times into the 12 Hour clock.

a. Train A _____

b. Train B _____

c. Train C _____

d. I arrived for a meeting and looked at the digital clock on the wall. It showed the correct time of 3:10pm. My watch was a few minutes slow and showed 2:56pm. How many minutes slow was my watch? _____

---

6. Work out the missing digits and write them in the spaces below.

a.

3 1 3
× ___

= 1 2 5 2

b.

__ 1 __
× 3

= 9 __ 2

c.

8 8 __
÷ 2

= __ __ 1

d.

__ __ 9
÷ 3

= 2 1 __

---

7. Complete the following ratio questions.

a. In Joseph's office the ratio of men to women is 2:5. If there are 20 women in the office how many men are there? _____

b. In Paul's office the ration of men to women is 3:1. If there are 18 men in the office how many women are there? _____

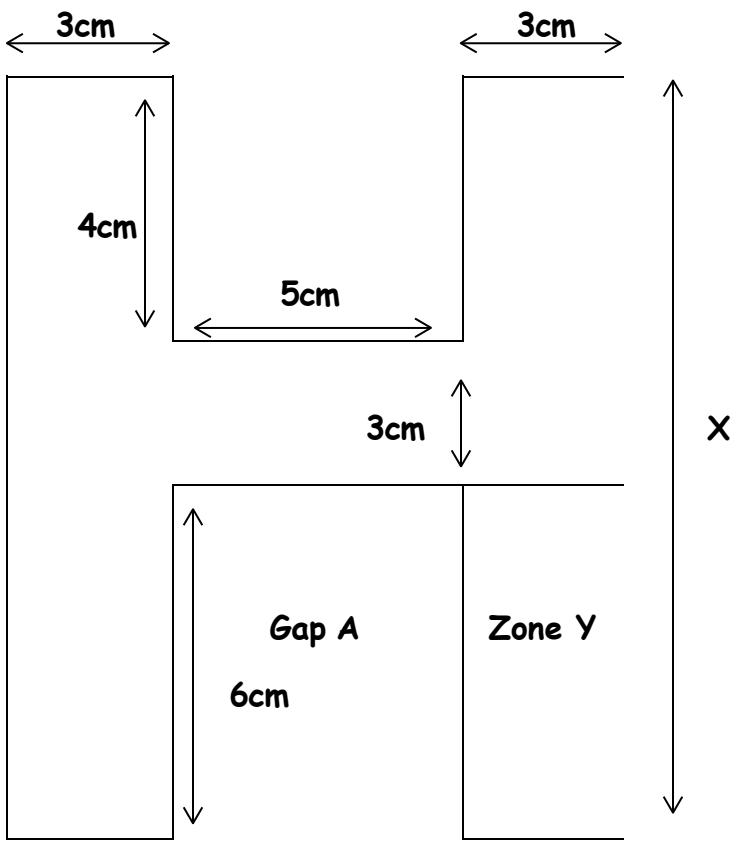
c. In Joanna's office the ratio of men to women is 6:3. If there are 15 women in the office how many men and women combined are there?

_____

d. If office A holds 36 people in a ratio of men to women of 8:1, how many men would there be and how many women?

Men _____ Women _____

8. Study this shape then answer the questions below.



a. How long is the side marked X? _____

b. What area in cm^2 is Gap A? _____

c. Calculate the total area of the shape. _____

d. Calculate the perimeter of the shape. _____

e. What is the area of zone Y? _____

---

9. Carry out the following measurement conversions.

a. Change 10cm to millimetres.

_____

b. Change 6.5km to metres.

_____

c. Change 500ml to centilitres

_____

d. Change 150cl to litres.

_____

e. Change 4.25kg to grams.

_____

f. Change 29 metres to centimetres. _____

g. Change 5 litres to millilitres.

_____

h. Change 3750g to kilograms.

_____

10. Look at the following statements and indicate if they are TRUE or FALSE.

a. The interior angles of any quadrilateral always add up to 360°. _____

b. A heptagon has seven sides. _____

c. The internal angles of a triangle add up to 180°. _____

d. A pentagon has six sides. _____

e. The area of a triangle is the product of its base and its vertical height. _____

f. A decagon has ten sides. _____

g. The interior angles in an equilateral triangle are 80° each. _____

---

11. Calculate the value of the letter in each of the following.

a. 4a + 8 = 32 a = _____

b. 3b = 18 b = _____

c. 13 – c = 8 c = _____

d. d + 25 = 37 d = _____

e. 56 ÷ 7 = e e = _____

f. (88÷f) – 13 = 9 f = _____

---

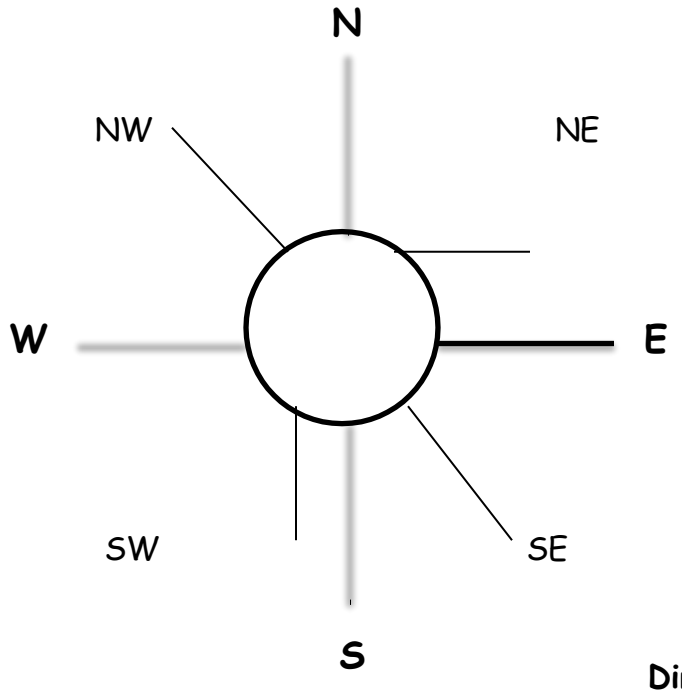
12. In a sale all items are marked down by 10%. Calculate what the sale price of the following items would be.

a. A skirt £30.00 __ sale price _____

b. A blazer £85.00 __ sale price _____

c. Trousers £45.00 __ sale price _____

13. The diagram below represents a compass. Imagine you are standing in the middle of it and then answer the following questions.



Direction facing at end of manoeuvre

a. I start facing South and turn clockwise through 270°. _____

b. I start facing West and turn clockwise through 90°. _____

c. I start facing North and turn anti-clockwise through 270°. _____

d. I start facing East and turn anti-clockwise through 135°. _____

e. I start facing South and turn clockwise through 180°. _____

f. I start facing West and turn anti-clockwise through 90°. _____

END OF TEST

If you have time

- **Go back and complete any questions you missed out.**
- **Check your answers carefully.**

Mathematics Revision Test 26

As you only have 10 questions I expect full marks please!

1. $\frac{2}{3}$ of 75p = _____

1. (27 ÷ 3) = (_____ ÷ 7)
   The two sides must balance in this sum.

2. Write as a percentage:
   $\frac{63}{100}$ = _____% b) 0.07 = _____%

3. $\frac{2kg\ 400g}{5}$ = _____ g

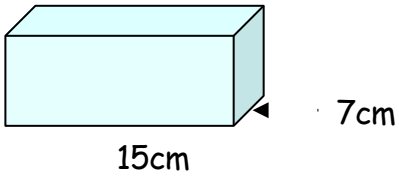
4. How many right angles are there in 270°? _____

5. 0.35 = _____ thousandths

6. A QUADRILATERAL with one pair of parallel lines is called a _____.

7. a) How many cm cubes fit exactly into the first layer of the box?
   _____

   a) If the box is 3 cm high, how many cm cubes are needed to fill it? _____



8. £70 ÷ 20 = _____

9. 350 ml x 8 = _____ litres

If you get any wrong please go over any corrections.

Fractions - reducing to lowest terms

Look at this code:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$\frac{1}{2}$	$\frac{2}{5}$	$\frac{5}{6}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{9}$	$\frac{3}{5}$	$\frac{5}{9}$	$\frac{1}{3}$	$\frac{7}{10}$	$\frac{1}{10}$	$\frac{1}{7}$	$\frac{2}{9}$	$\frac{3}{8}$	$\frac{3}{4}$	$\frac{3}{10}$	$\frac{5}{7}$	$\frac{5}{8}$	$\frac{2}{3}$	$\frac{4}{5}$	$\frac{1}{6}$	$\frac{2}{7}$	$\frac{1}{5}$	$\frac{7}{8}$	$\frac{3}{7}$	$\frac{4}{9}$

By reducing each of the following fractions to its lowest terms you can find out which letter it represents.

For example $\frac{3}{6}$ can be reduced to $\frac{1}{2}$ and so it represents A

Write the appropriate letter <u>above</u> each of the fractions below and discover the hidden messages.

1. $\frac{8}{10}$ $\frac{10}{18}$ $\frac{3}{9}$ $\frac{12}{18}$ $\frac{10}{12}$ $\frac{6}{8}$ $\frac{3}{24}$ $\frac{3}{12}$ $\frac{2}{6}$ $\frac{8}{12}$ $\frac{3}{12}$ $\frac{5}{10}$ $\frac{6}{9}$ $\frac{6}{14}$

2. $\frac{4}{18}$ $\frac{4}{8}$ $\frac{12}{15}$ $\frac{10}{18}$ $\frac{4}{6}$ $\frac{15}{18}$ $\frac{2}{4}$ $\frac{9}{24}$ $\frac{6}{15}$ $\frac{2}{8}$ $\frac{2}{18}$ $\frac{2}{12}$ $\frac{6}{16}$

3. $\frac{9}{24}$ $\frac{9}{12}$ $\frac{3}{15}$ $\frac{6}{14}$ $\frac{15}{20}$ $\frac{3}{18}$ $\frac{10}{20}$ $\frac{15}{24}$ $\frac{5}{20}$ $\frac{6}{10}$ $\frac{3}{12}$ $\frac{12}{15}$ $\frac{8}{10}$ $\frac{4}{12}$ $\frac{6}{16}$ $\frac{9}{15}$ $\frac{3}{27}$ $\frac{5}{10}$ $\frac{8}{12}$ $\frac{16}{20}$ $\frac{3}{12}$ $\frac{10}{16}$

4. $\frac{12}{15}$ $\frac{15}{27}$ $\frac{5}{20}$ $\frac{9}{24}$ $\frac{2}{8}$ $\frac{21}{24}$ $\frac{16}{20}$ $\frac{2}{18}$ $\frac{3}{12}$ $\frac{3}{15}$ $\frac{12}{24}$ $\frac{15}{24}$ $\frac{4}{16}$ $\frac{10}{18}$ $\frac{6}{12}$ $\frac{10}{16}$ $\frac{3}{24}$ $\frac{2}{8}$ $\frac{15}{24}$

5. $\frac{24}{30}$ $\frac{15}{27}$ $\frac{4}{12}$ $\frac{20}{30}$ $\frac{8}{40}$ $\frac{27}{36}$ $\frac{20}{32}$ $\frac{3}{30}$ $\frac{4}{20}$ $\frac{4}{12}$ $\frac{5}{35}$ $\frac{2}{14}$ $\frac{10}{18}$ $\frac{6}{24}$ $\frac{3}{21}$ $\frac{6}{20}$ $\frac{6}{14}$ $\frac{12}{16}$ $\frac{5}{30}$

6. $\frac{8}{40}$ $\frac{8}{32}$ $\frac{6}{42}$ $\frac{4}{28}$ $\frac{4}{32}$ $\frac{6}{8}$ $\frac{12}{32}$ $\frac{5}{20}$

Maths 11 Assessment

Test

The test paper attached is an assessment for 10 and 11 year olds. This paper will be a good indicator of the sort of Maths questions you are likely to get in an 11+ exam.

Please find time during the next 2 weeks to complete this assessment test. There is no time limit, so don't rush, and check your answers carefully.

Don't worry if you have problems with any of the questions, because we have from now until the exam to put right any problem areas. Our Maths course is not finished yet!

Good luck!

Mathematics 11 (Alternative)

Assessment Test

Name _____

School _____ Class _____

Today's date _____

Date of Birth _____

Note

- No help from parents allowed!
- There is no time limit.
- Do not rush your work.
- Check over your work carefully.
- Every correct answer counts.
- No calculators please.

1.

$$\begin{array}{r} 314 \\ +\ 689 \\ \hline \\ \hline \end{array}$$

---

2. Write 7049 in words:

Answer _____

---

3. Look at this list of lengths:

50cm 160cm $1\frac{1}{2}$ m 0.8m 100cm

Write these lengths in order of size.

Put the shortest length first.

Answer _____

_____

_____

_____

_____

---

4. Put a tick by the time which is the same as 9:35 pm.

9:35 am	
25 to 10	
half past 9	
9:40	
quarter to 9	

5.

$$\begin{array}{r} 389 \\ \times\ 4 \\ \hline \\ \hline \end{array}$$

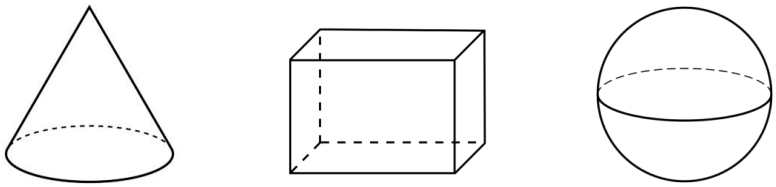
6.

$$\begin{array}{r} 878 \\ -\ 289 \\ \hline \\ \hline \end{array}$$

7.

A cone
B triangular prism
C sphere
D cuboid

Put the correct letter under each shape.



_____ _____ _____

8. Four pieces of carpet are cut from a roll 2 metres long.
   Each piece is 40cm long.

   How long is the piece left over?

   Answer _____ cm

---

9. Jeff has 77p.
   He has **4** coins.

   Which coins are they?

   Answer _____p

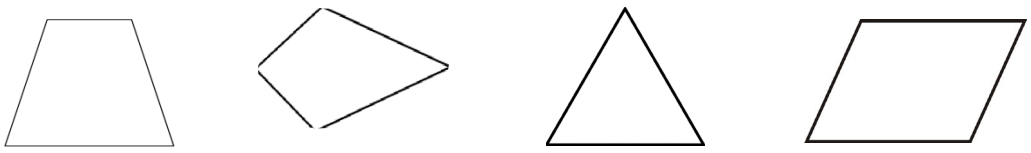
   _____p

   _____p

   _____p

---

10. Which of these is **not** a quadrilateral?



A **B** **C** **D**

Answer _____

---

11. Put the correct number in the box.

$$34 \times 101 = 3400 + \boxed{}$$

Look at the numbers in the box.

	5.8	0.076	17.7	0.375
5.9				
		0.92	1.25	
	97.2	1.6		67
			3.87	

12. Which is the smallest of these numbers?

Answer _____

13. How many of these numbers are smaller than 5?

Answer _____

---

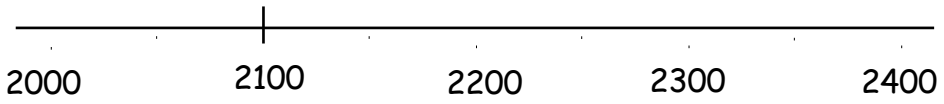
14. Jessie wants to buy a bicycle.
She has saved £48.80.
The bicycle costs £65.

How much more money does she need?

Answer £ _____

---

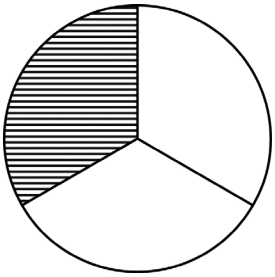
15. Mark with a cross the number 2175.



16. Write 0.25 as a fraction.

Answer _____

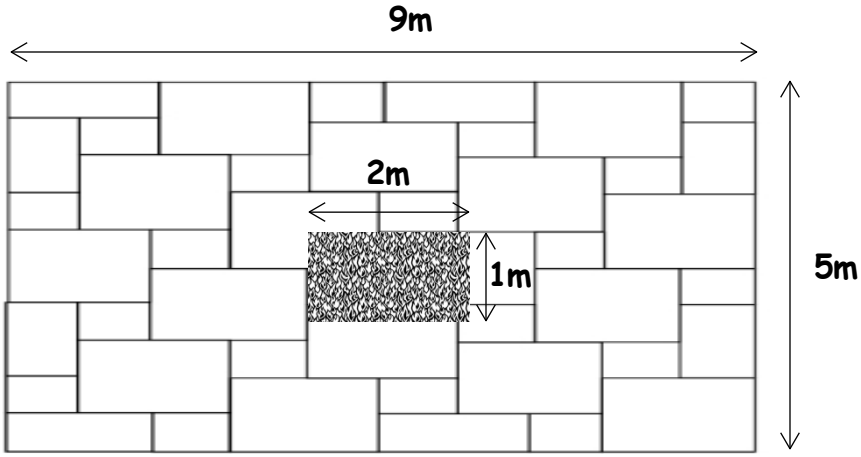
17.



What fraction of the circle is shaded?

Answer _____

18.



Will's patio was 9m long and 5m wide.

He decided to add a fireplace measuring 2m long and 1m wide.

What area of lawn was left?

Answer _____

19. What percentage of £2 is 2p?

Answer _____

---

20. 350 divided by 5 = _____

---

21. Write the correct number in each box.

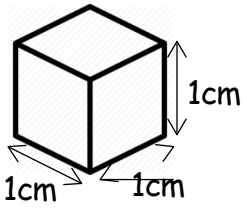
456 + ☐ = 456

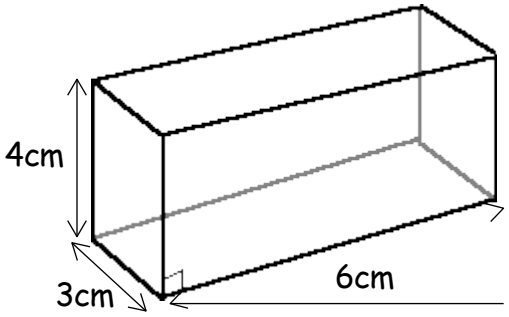
456 - ☐ = 0

456 x ☐ = 0

456 ÷ ☐ = 1

22.





How many 1cm cubes will fit in the container?

Answer _____ cubes

23. 5.27 + 0.126 = _____

---

24. Subtract 36 from 79.

Answer _____

25.

17 31 59

The three numbers above are alike in some ways.

Underline two ways in which they are alike.

They are all even number.

They can all be divided exactly by 3.

They are all two-figure numbers.

They are all prime numbers.

They are all square numbers.

They can all be divided exactly by 2.

---

26. The centre of a circle is 30cm away from the circumference.

What is the diameter of the circle?

Answer _____ cm

27. The time in Tokyo is 8 hours ahead of the time in London.

In London it is 10am.

What time is it in Tokyo?

Answer _____

---

28. How many sides has a hexagon?

Answer _____

---

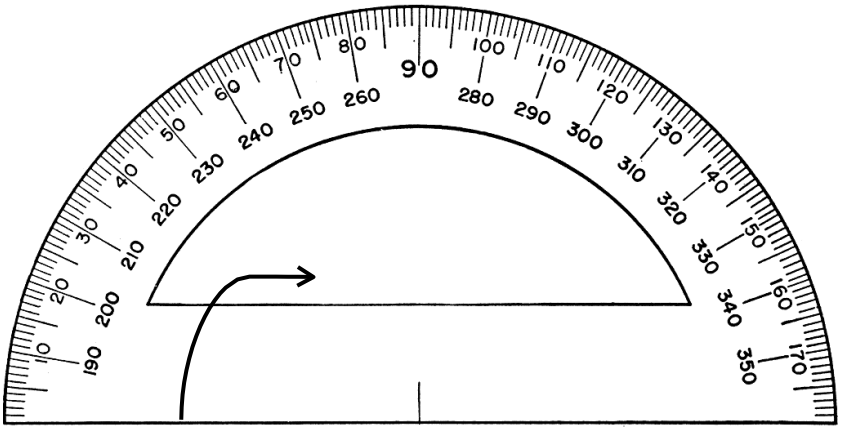
29. On a map a river is 2 $\frac{1}{4}$ centimetres long.

The scale of the map is 100 miles to 1 centimetre.

How long is the road?

Answer _____

30.



How many degrees does this angle measure?

Answer _____ degrees

---

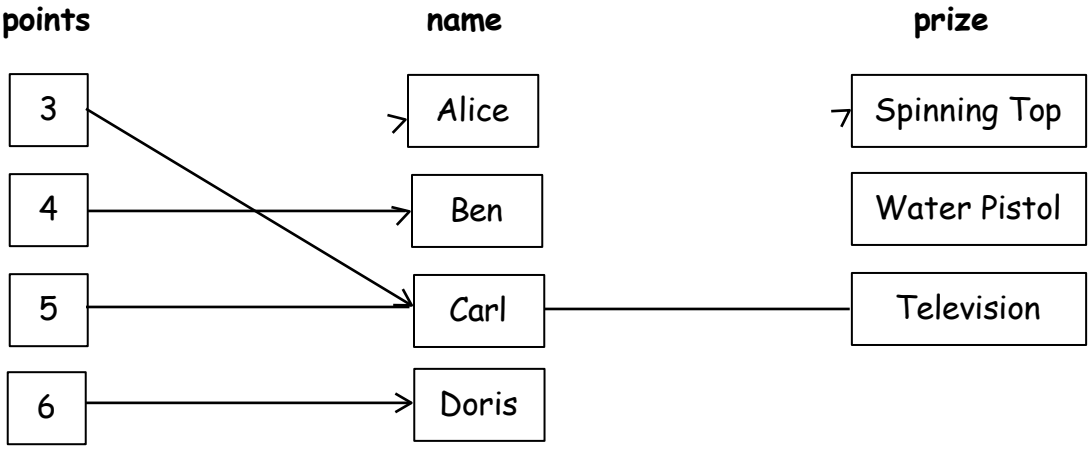
31. Shade in 3/5 of this shape.

---

32. Look at the table.

points	prize
0-3	Spinning Top
4-5	Water Pistol
6-7	Television

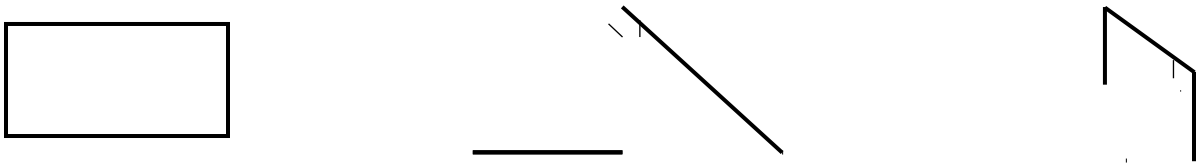
Now complete the mapping.



33. Write $\frac{1}{4}$ in decimal form

Answer _____

34.



These shapes are alike in some ways.

Underline two ways in which they are alike

They all have 4 sides.

They are all rectangles.

They all have the same area.

They all have at least one right angle.

They are all triangular.

---

35. 1.37 x _____ = 137

---

36. This chart shows Johnny's activities for 8 hours on one day.

How many hours was he sleeping?

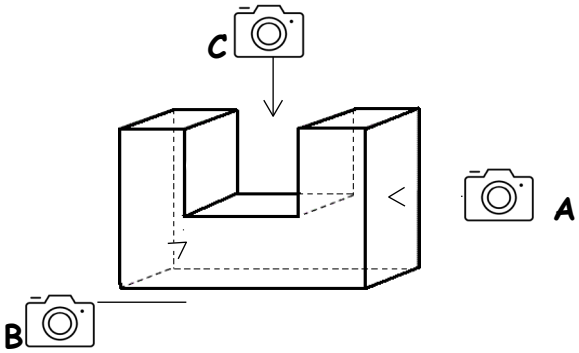
Answer _____ hours

---

37. 1386 ÷ 6 = _____

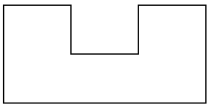
---

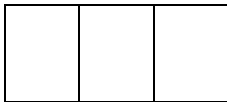
38. Look at this picture of a block of wood.



A view was taken by each camera.

Put the correct letter under each view





____ ____ ____

39. My average speed is 40 miles per hour and I drive for 7 hours.

How many miles have I travelled?

Answer _____ miles

---

40. This chart shows how long Peter took to shower each day last week.

Day	**Minutes**
Monday	12 minutes
Tuesday	23 minutes
Wednesday	5 minutes
Thursday	11 minutes
Friday	18 minutes

What was the average time taken?

Answer _____ minutes

---

41. Ewan saves £12.50 a month.

How many months will it take for Ewan to save £50?

Answer _____ months

---

42. Put the correct number in the box.

1/3 hour = ☐ minutes

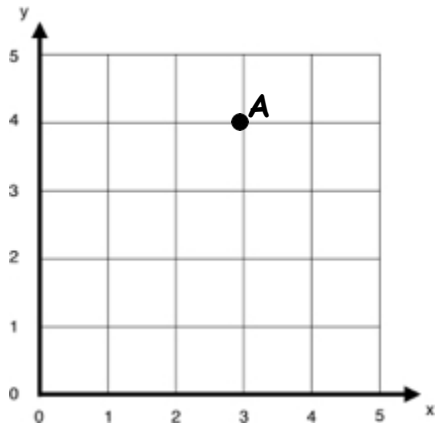
43. A water bottle weighs 200 grams.

When it is filled with water it weighs 1 kilogram 500 grams.

How much does the water weigh?

Answer _____ grams

44.



What are the co-ordinates of A?

Answer _____

---

45. What is 70% of 50?

Answer _____

---

46. One side of Mr Brown's roof measures 12m by 3m.

Solar panels measure $\frac{1}{2}$m by $\frac{1}{2}$m.

How many solar panels can Mr Brown fit on this one side of the roof?

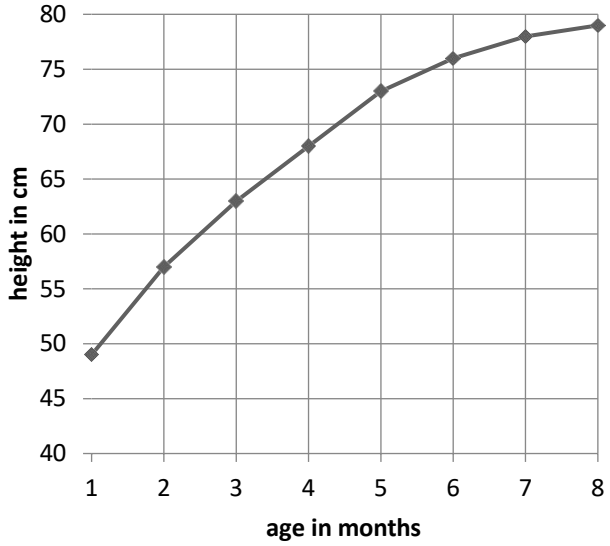
Answer _____

47. The length of each edge of a cube was halved.

How much smaller did the volume become?

Answer _____

The graph shows the height of a baby boy over the first eight months of his life



48. What was the baby's height at birth?

Answer _____ cm

49. How tall was the baby after 7 months?

Answer _____ cm

50. How tall was the baby at the end of the month in which he grew the most?

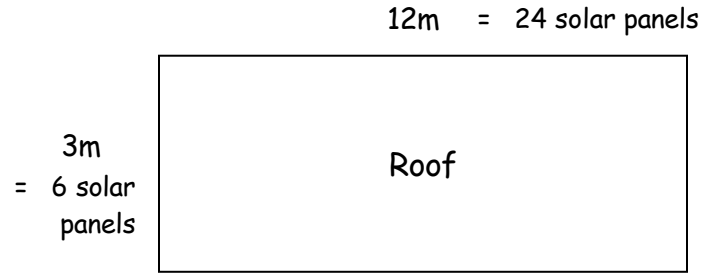
Answer _____ cm

Most children find these two problems impossible!
Have a look at them again.

Question 46

One side of Mr Brown's roof measures 12m by 3m.
Solar panels measure $\frac{1}{2}$ m by $\frac{1}{2}$ m.
How many solar panels can Mr Brown fit on this one side of the roof?

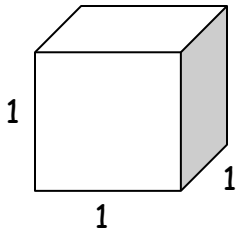
The side of each solar panel measures $\frac{1}{2}$m, so 24 solar panels are needed for 12m, and 6 solar panels for 3m. **24 x 6** = **144**. This is the answer.
Not that hard really, but everyone finds this tricky, so if you do, you are not alone! Draw a diagram if this helps like the one below:



Question 47

The length of each edge of a cube was halved.
*How much smaller did the **volume** become?*

Look at a cube which is 1cm on each edge. The volume of this cube is 1x1x1, which is 1cm^3 (1 cubic cm)



If the length of each edge is **halved** then the measurement would be $\frac{1}{2}$x$\frac{1}{2}$x$\frac{1}{2}$, making the volume $\frac{1}{8}$ cm^3. Therefore, the volume is 8 times smaller.

Try again:
Cube with sides of 4cm = 4x4x4 = 64cm^3
Halve this measurement = 2x2x2 = 8cm^3
8 is **8 times smaller** than 64.

Calculation Ladders

Note: n^2 means the number has to be squared, for example $6^2 = 36$

You may either complete them in your head, or fill in your answers at the side as you work down the ladders.

Follow the instructions to the bottom, where you write in your answer.

Ladder 1

1.	9
2.	X 4
3.	Take away 8
4.	÷4
5.	Subtract 3
6.	n^2
7.	halve it
8.	x 6
9.	÷ 12
10.	Multiply by three
	Answer:

Ladder 2

1.	20
2.	Multiply by 3
3.	Half of this
4.	Divide by 10
5.	11 more
6	Share equally between two
7.	X 3
8.	Plus 7
9.	$\frac{1}{2}$ of this
10.	$\frac{1}{2}$ of this
	Answer:

Calculation Ladders

Everyone loves this work on percentages so I hope you do too!

Ladder 3

1.	**£25**
2.	÷5
3.	x 4
4.	Double it
5.	Divide by 8
6.	n^2
7.	Plus £3
8.	Share into 7 equal parts
9.	x 4
10.	Add £5
	Answer:

Ladder 4

1.	**18**
2.	Add 2
3.	÷5
4.	Add 8
5.	Double it
6.	A quarter of this
7.	-3
8.	Times seven
9.	Double this number
10.	÷6
	Answer:

ANSWERS - COMPLETE MATHS – PART 28

Maths

Maths type 1 paper 3

Page 1

1.
a. 70%
b. 26%
c. 80%
d. 12%
e.Fergus and Nadia

2.
a. 5/8
b. 5/12
c. 1/8
d. 19/20
e. 10.4
f. 0.54
g. 4.06
h. 12.4
i. 21:15
j. 6:30am
k. 16:25

Page 2

3.
a. 0
b. 5
c. 2
d. 6
e. 8
f. 1

4.
a. 12
b. 56
c. 35
d. 24

Page 3

5.
a. 9:30am
b. 11:45am
c. 3:15pm
d. 14 mins

6.
a. 313
x 4
= 1252

b. 3 1 4
x 3
= 9 4 2

c. 882
÷ 2
= 441

d. 639
÷ 3
= 213

7.
a. 8
b. 6
c. 30
d.32& 4

Page 4

8.
a. 13cm
b. 30cm²
c. 93cm²
d. 68cm
e. 18cm²

9.
a. 100mm
b. 6500m
c. 50cl
d. 1.5l
e. 4250g
f. 2900cm
g. 5000ml
h. 3.75kg

Page 5

10.
a. true
b. true
c. true
d. false
e. false
f. true
g. false

11.
a. 6
b. 6
c. 5
d. 12
e. 8
f. 4

12.
a. £27
b. £76.50
c. £40.50

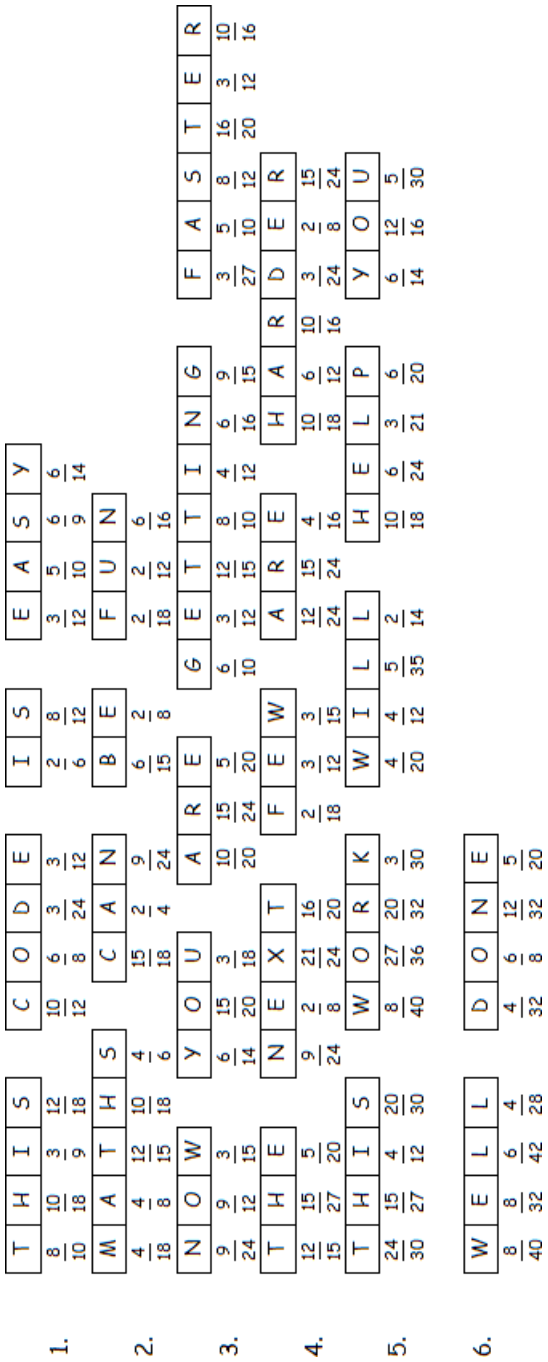
Page 6

13.
a. E
b. N
c. E
d. NW
e. N
f. S

Revision sheet

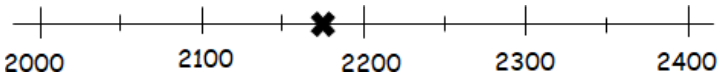
1. 50p
2. 63
3. 63% 7%
4. 480g
5. 3
6. 350
7. Trapezium
8. 105 315
9. £3.50
10. 2.8L

Fractions



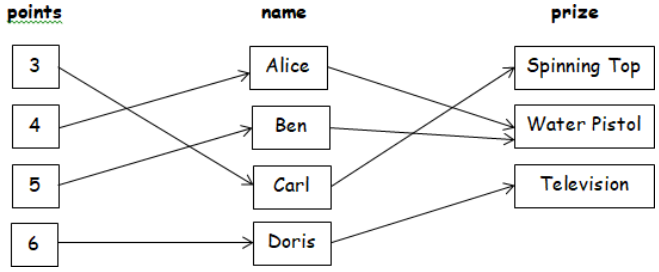
Maths 11 Assessment

1. 1003
2. seven thousand and forty nine
3. 50cm, 0.8m, 100cm, 1 $\frac{1}{2}$ m, 160cm
4. 25 to 10
5. 1556
6. 589
7. A, D, C
8. 40cm
9. 50p, 20p, 5p, 2p
10. C
11. 34
12. 0.076
13. 6
14. £16.20
15.
16. $\frac{1}{4}$
17. 1/3



18. 43m^2
19. 1%
20. 70
21. 0, 456, 0, 456
22. 72
23. 5.396
24. 43
25. They are all two-figure numbers & they are all prime numbers.
26. 60cm
27. 6pm (18:00)
28. 6
29. 225 miles
30. 55 degrees
31.

32.



33. 0.25

34. They all have 4 sides, they all have at least one right angle
35. 100
36. 3 hours
37. 231
38. B, C, A
39. 280 miles
40. 13.8 minutes (13 minutes 48 seconds)
41. 4 months
42. 20 minutes
43. 1300 grams
44. (3, 4)
45. 35
46. 144 solar panels
47. 8 times smaller
48. 49 cm (allow 1cm either side for error in measurement)
49. 78 cm (allow 1cm either side for error in measurement)
50. 57 cm (allow 1cm either side for error in measurement)

Calculation Ladders

- Test 1 – 12
- Test 2 - 7
- Test 3 – £21
- Test 4 - 7

(Be careful when you are dealing with money)